NetExam

Sri Lanka Institute of Information Technology

Which one is the true statement regarding systems with web-services?

Select one:

- a. Supports dynamic integration of components
- b. Use proprietary protocols for communication
- c. Web services always use monolithic architecture
- d. Web services are about user to program interaction

NetExam

Sri Lanka Institute of Information Technology

Which language supports both procedural and object oriented programming?

Select one:

- a. Java
- b. C
- c. C++
- d. C#

oop- C++, Python, Java

procedural - C++ , Python

NetExam

Sri Lanka Institute of Information Technology

The CRUD operations are determined by?

1. Read & Create
2. PUT & DELETE
3. HTTP verb
4. All of them

HTTP verbs (or methods, as they are properly called) are POST, GET, PUT, PATCH, and DELETE.

These correspond to create, read, update, and delete (or CRUD) operations

PUT - creates a new resource or replaces a representation of the target resource with the request payload.

Select one:

- a. Only (3)
- b. Only (4)
- c. Only (2)
- d. Only (1) and (2)

NetExam

Sri Lanka Institute of Information Technology

Which of the following is not a software category?

Select one:

- a. CASE tools
- b. Utilities
- c. System software
- d. Cloud computing

Cloud computing is the service, not a software

NetExam

Sri Lanka Institute of Information Technology

Identify correct ways to maintain the quality of the code:

1. Documenting the code
2. Functional testing
3. Regular code reviews
4. Clear requirements

Select one:

- a. Only (2) and (3)
- b. Only (1) and (3)
- c. Only (2) and (4)
- d. All (1), (2), (3), and (4)

Evaluate WSDL web service description language

Select one:

- a. All of others
- b. WSDL helps in dynamic service integration
- c. WSDL is a contract between the Web service and the client
- d. WSDL describes a web service

NetExam

Sri Lanka Institute of Information Technology

What is the verb, which we use to indicate Create operation? (1) GET (2) POST (3) PUT (4) DELETE

Select one:

- a. Only (2)
- b. Both of them
- c. None of them
- d. Only (3)

POST =>> Create
GET =>> Read
PUT =>>Update/Replace
PATCH =>> Update or Modify
DELETE =>> Delete

NetExam

Sri Lanka Institute of Information Technology

What is not related to the scalability?

Select one:

- a. Distribution of the components
- b. Testing
- c. Number of users
- d. Concurrent access

NetExam

Sri Lanka Institute of Information Technology

Some applications developed using particular programming languages can be directly run on the OS witho

Select one:

- a. This statement is true
- b. Cannot validate the statement without knowing the OS
- c. This statement is not conclusive
- d. This statement is false

NetExam

Sri Lanka Institute of Information Technology

What is false about Micro-services Architecture?

Select one:

- a. The system is developed with many components
- b. Can use single database or multiple databases
- c. May use SOAP or REST for communication
- d. Made of many web services

components:

Clients
Identity Providers
API Gateway
Messaging Formats
Databases
Static Content
Management
Service Discovery

NetExam

Sri Lanka Institute of Information Technology

What are the languages, which can be used to write code in an HTML page, which can

Select one:

- a. CSS, Java
- b. None of the above
- c. JS, Java
- d. CSS, JS

NetExam

Sri Lanka Institute of Information Technology

The communication techniques in distributed systems can be categorized as

Select one:

- a. RPC, RMI, and CORBA
- b. Functional oriented, Message oriented, and Resource oriented
- c. RPC/RMI, SOAP, and DC
- d. Delta-Communication, SOAP communication, and RESTful communication

Functional oriented communication
• RPC/RMI
• CORBA

2. Message Oriented communication
• SOAP

3. Resource oriented communication
• REST

4. Communication in Rich Web-based Applications

NetExam

Sri Lanka Institute of Information Technology

Refactoring is one of the aspect of quality code. What does it mean?

Select one:

- a. Re-arrange the code to improve efficiency
- b. Manage the configurations
- c. Extending the code
- d. Test and fix the bugs

NetExam

Sri Lanka Institute of Information Technology

Web-based applications can be categorized into

Select one:

- a. standalone applications and client-server applications
- b. standalone applications and browser-based applications
- c. browser-based applications and non-browser-based applications
- d. web applications and browser-based applications

brower base / non browser base

or

client server application / web application

NetExam

Sri Lanka Institute of Information Technology

What are the benefits of web services?

1. Static integration
2. Standardized Protocol
3. Interoperability
4. Constant by design
5. Low Cost of communication

Select one:

- a. 1 and 5
- b. 3, 4, and 5
- c. 2, 3, and 5
- d. 1, 2, and 4

Interoperability has the highest priority.

NetExam

Sri Lanka Institute of Information Technology

Procedural programming focuses on

Select one:

- a. Flow control with GOTO statements
- b. Use of namespaces/packages
- c. Functions (as in functional programming)
- d. Modularity and code reuse

NetExam

Sri Lanka Institute of Information Technology

Which cannot be used to develop the server-model?

Select one:

- a. PHP
- b. HTML
- c. Node.JS
- d. JAVA

•To develop the server model, a server-side language/framework and related technologies should be used.

•PHP, JAVA, .NET, Python,

Most large-scale websites use server-side code to dynamically display different data when needed, generally pulled out of a database stored on a server and sent to the client to be displayed via some code (e.g. HTML and JavaScript).

NetExam

Sri Lanka Institute of Information Technology

The difference between the Servlets and the JSP:

Select one:

- a. Servlets are used to develop web applications and JSP is used to develop web sites
- b. JSP is an application and Servlet is a web page
- c. Servlet can generate HTML and JSP can generate JAVA code
- d. JSP is a web page and Servlet is an application

JSP – Web pages with Java code
•Servlets – Java applications, which generate HTML

Server-side - frameworks and libraries
• JAVA – JSP/Servlet

NetExam

Sri Lanka Institute of Information Technology

Which one of the following is not based on GIT?

Select one:

- a. SourceForge
- b. GitHub
- c. OneDrive
- d. GitLab

gitBucket
GitHub
GitLab
SourceForge
cloudForge

NetExam

Sri Lanka Institute of Information Technology

The resource is determined by? (1) The domain name (2) CRUD operations (3) URL segments

Select one:

- a. None of them
- b. Only (3)
- c. Only (2)
- d. Only (1)

NetExam

Sri Lanka Institute of Information Technology

Which one of the following is not a benefit of CDN? Content Delivery Network

Select one:

- a. Reducing bandwidth costs
- b. Improving the security of the system
- c. Support collaboration
- d. Improving system's loading time

Increasing content available and redundancy

NetExam

Sri Lanka Institute of Information Technology

What modules are implemented on the server-side of BAW-MVC?

1. Component
2. Controller
3. View
4. Model

Select one:

- a. Only (1) and (2)
- b. Only (1), (2), and (3)
- c. Only (3), and (4)
- d. All of them

NetExam

Sri Lanka Institute of Information Technology

What are the advantages of MVC?

Select one:

- a. Presentation is separated from the business logic
- b. All of the others
- c. Event-handling is separated from business logic
- d. Presentation is separated from Event-handling

NetExam

Sri Lanka Institute of Information Technology

Programming languages provide

Select one:

- a. Semantics and functions
- b. Syntax and procedures
- c. Syntax and functions
- d. Syntax and semantics

Bootstrap is a?

Select one:

- a. Scripting language
- b. Plugin
- c. Framework
- d. Library

NetExam

Sri Lanka Institute of Information Technology

CSS can be written as

Select one:

- a. all of other answers
- b. Inline CSS
- c. External CSS sheets
- d. Internal CSS sheets

NetExam

Sri Lanka Institute of Information Technology

Which one of the following is not a VCS model?

Select one:

2 a. Centralized version control systems

1 b. Local version control systems

3 c. Distributed Version Control Systems

d. Global version control systems

Can be used for collaborative software development

Oldest VCS

• Everything is in your & Computer Cannot be used for collaborative software developmen

• No single point of failure

Why do centralized control in distributed systems useful to engineers?

Select one:

- a. Maintaining the system is easier
- b. All of the others
- c. Easier to update
- d. Easier to modify and extend

NetExam

Sri Lanka Institute of Information Technology

What is NOT a correct statement about CSS?

Select one:

- a. None of the others
- b. Eliminate HTML formatting tags
- c. Save work and time by reusing style sheets
- d. Slow down loading

CSS is Fast Loading

NetExam

Sri Lanka Institute of Information Technology

What is the operation for HTTP verb PUT? (1) Create (2) Update

Select one:

- a. None of them
- b. Only (1)
- c. Only (2)
- d. Both of them

POST =>> Create
GET =>> Read
PUT =>>Update/Replace
PATCH =>> Update or Modify
DELETE =>> Delete

NetExam

Sri Lanka Institute of Information Technology

Which of the following is incorrect about JAX-RS annotations?

Select one:

a. PATH - Relative path of the resource class/method

b. POST - Create a new resource

c. PUT - Used to add resources

d. HEAD - Get status about method availability

•JAX-RS is an API
•There are two implementations

jersey

RESTeasy

NetExam

Sri Lanka Institute of Information Technology

Which of the following is not correctly defining loosely coupled architecture of web service?

Select one:

- a. services can be independently updated
- b. software systems become more manageable
- c. allows integration between different systems
- d. Low scalability

NetExam

Sri Lanka Institute of Information Technology

"In software runtime architecture we may need additional virtualization layers, depending on the type of the system." Between which layers we may need these virtualization layers?

Select one:

- a. Between Operating System and Hardware
- b. Between User and Application
- c. Between User and Hardware
- d. Between Application and Operating System

NetExam

Sri Lanka Institute of Information Technology

the Model-View-Controller (MVC) architectural style, the model defines the

elect one:

- a. Interfaces
- b. Presentation
- c. Business-logic
- d. Communication model

NetExam

Sri Lanka Institute of Information Technology

The key technique in RESTful communication uses: (1) URL (2) HTTP Verbs

Select one:

- a. Only (1)
- b. None of them
- c. Only (2)
- d. Both of them

URL is the key technique in RESTful communication
• The API of the RESTful service is a set of URLs

*There is a gap between development and implementation environments, such as having different platfo

Which one of the following helps us to overcome these issues?

Select one:

- a. Version controlling
- b. CDN
- c. Dependency management
- d. Virtualization

NetExam

Sri Lanka Institute of Information Technology

What are the example for data representational elements in HTML?

Select one:

- a. Tables, Input
- b. Lists, Tables
- c. Input, Buttons
- d. Buttons, Tables

•Data representational elements
(these elements use nested structures)
• Lists
• Tables

•>>Form elements
• Input
• Radio buttons, check boxes
• Buttons

NetExam

Sri Lanka Institute of Information Technology

What is used by Maven to store project details

Select one:

- a. CDN
- b. VCS
- c. XML file
- d. JSON file

Which is NOT correct about the View development in web-based applications?

Select one:

- a. View is deployed to the client's computer
- b. View should be loaded to the client's browser
- c. Server-side code in View may execute in server-side
- d. View should communicate with model to load the data from database

NetExam

Sri Lanka Institute of Information Technology

Which one of the following is not a purpose of having a VCS?

Select one:

- a. Restoring previous versions
- b. Backup
- c. Code formatting
- d. Collaboration

Collaboration
• With a VCS, everybody on the team is able to work absolutely freely - on any file at any time
• The VCS will later allow you to merge all the changes into a common version

• Storing versions properly
• A version control system acknowledges that there is only one project

• Restoring previous versions

• Understanding what happened
• Every time you save a new version of your project, your VCS requires you to provide a short description of what was changed

• Backup

on 13

t answered

d out of

g question

The basic architecture of the distributed systems is called

Select one:

- a. Client-server architecture
- b. N-tier architecture
- c. 3-tier architecture
- d. Service oriented architecture

two tier

The basic architecture of the distributed systems is called the client-server (or two-tier) architectur

Sri Lanka Institute of Information Technology

What HTTP verb is used for Read operation?

(1) GET

(2) POST

(3) PUT

(4) DELETE

POST =>> Create
GET =>> Read
PUT =>>Update/Replace
PATCH =>> Update or Modify
DELETE =>> Delete

Select one:

- a. (1) and (2) only
- b. All of them
- c. Only (1)
- d. only (2)

NetExam

Sri Lanka Institute of Information Technology

A framework is incomplete?

Select one:

- a. The correct answer is not given
- b. Not conclusive
- c. Agree
- d. Disagree

A Framework is incomplete

- Framework itself is not usable on its own. Also they do not leave some fill-in-the-blanks for its user.
- Framework alone will not work, the relevant application logic should be implemented and be deployed along with the framework
- Frameworks pose trade-off challenge between learning curve and saving-coding-time

Bootstrap is a?

Select one:

- a. Scripting language
- b. Plugin
- c. Framework ✓
- d. Library

Scanned with CamScanner

What modules are implemented on the server-side of BAW-MVC?

1. Component
2. Controller ✓
3. View
4. Model ✓

No answer

Select one:

- a. Only (1) and (2)
- b. Only (1), (2), and (3)
- c. Only (3), and (4)
- d. All of them

Scanned with CamScanner

Question **21**

Answer saved

Marked out of 1.00

Flag question

A software framework is an abstraction in which software, providing generic functionality, can be selectively changed (extended) by additional user-written code, thus providing application-specific software. Do you agree with above statement?

Select one:

- a. Cannot say conclusively
- b. Agree.
- c. Disagree.
- d. The correct answer is not given
- e. Strongly disagree

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:14:53**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **23**

Answer saved

Marked out of 1.00

Flag question

Client-server architecture is one of the six constraints in REST. What is/ are true regarding this constraint?

Select one:

- a. It promotes low coupling between the client and the server.
- b. It promotes portability of the user interface.
- c. All of these statements are correct.
- d. It allows clients and servers grow independently.
- e. It improves scalability of the server components.

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:14:57**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **24**

Answer saved

Marked out of 1.00

Flag question

A merge conflict in Git is an event that occurs when,

Select one:

- a. A user tries to merge branches that have conflicting changes which git cannot resolve on its own.
- b. A user accidentally deletes a branch which was to be merged with another branch.
- c. A repository becomes corrupted and unusable, therefore preventing the users from merging any further commits.
- d. None of the above
- e. A user forgets to commit changes to a repository.

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:14:58**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **25**

Answer saved

Marked out of 1.00

Flag question

What is not an advantage of using software frameworks?

Select one:

- a. Community support
- b. Users of a framework should respect the limitations and conventions imposed by its design.
- c. Enables faster methods for development with less code
- d. Accelerated development
- e. All of the above.

limitation

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left 0:15:01

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **27**

Answer saved

Marked out of 1.00

Flag question

Which of the following is NOT an advantage of following a good Git Workflow?

Select one:

- a. Clarity in identifying the software development requirements.
- b. Faster and efficient development practices
- c. Easy team collaboration
- d. The correct answer is not given
- e. Easy source code management

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:15:06**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **28**

Answer saved

Marked out of 1.00

Flag question

Representational State Transfer Or REST is a/ an,

Select one:

- a. Architectural protocol for facilitating communications on the web.
- b. Web API protocol.
- c. Architectural style for facilitating communications on the web.
- d. Technology for creating web APIs.
- e. All of the above.

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:15:08**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **29**

Answer saved

Marked out of 1.00

Flag question

What is/are a disadvantage(s) of using software frameworks?

Select one or more:

- [x] a. None of the given.
- [ ] b. Learning the Framework and not the programming language prevent programmers from gaining an in-depth understanding of the programming language.
- [ ] c. Users of a framework are forced to respect the limitations and conventions imposed by its design.
- [ ] d. Community support.
- [ ] e. Compliance to accepted security standards.

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left 0:15:11

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **30**

Answer saved

Marked out of 1.00

Flag question

What is a "pull request" in Git, and how is it used in collaboration?

Select one:

- a. It is a request to merge changes from one branch into another and can be reviewed and discussed by other developers.
- b. It is a combination of git fetch and git merge commands.
- c. It is a request to delete a branch or repository and can be reviewed and discussed by other developers.
- d. It is a request to download changes from a remote repository and merge them into a local repository.
- e. It is a request to create a new branch or repository and can be reviewed and discussed by other developers.

Previous page

Finish attempt ...

Quiz navigation

Finish attempt ...

Time left **0:15:16**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **30**

Answer saved

Marked out of 1.00

Flag question

What is a "pull request" in Git, and how is it used in collaboration?

Select one:

- a. It is a request to merge changes from one branch into another and can be reviewed and discussed by other developers.
- b. It is a combination of git fetch and git merge commands.
- c. It is a request to delete a branch or repository and can be reviewed and discussed by other developers.
- d. It is a request to download changes from a remote repository and merge them into a local repository.
- e. It is a request to create a new branch or repository and can be reviewed and discussed by other developers.

Previous page

Finish attempt ...

Quiz navigation

Finish attempt ...

Time left 0:15:16

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **30**

Answer saved

Marked out of 1.00

Flag question

What is a "pull request" in Git, and how is it used in collaboration?

Select one:

- a. It is a request to merge changes from one branch into another and can be reviewed and discussed by other developers.
- b. It is a combination of git fetch and git merge commands.
- c. It is a request to delete a branch or repository and can be reviewed and discussed by other developers.
- d. It is a request to download changes from a remote repository and merge them into a local repository.
- e. It is a request to create a new branch or repository and can be reviewed and discussed by other developers.

Previous page

Finish attempt ...

Quiz navigation

Finish attempt ...

Time left **0:15:16**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **3**

Answer saved

Marked out of 1.00

Flag question

What is/ are not an architectural constraint of REST?

1. Client-server architecture
2. Stateless
3. Cacheable
4. Strong Decoupling
5. Layered system

Select one:

- a. All of the above.
- b. 1, 2, 3 only.
- c. 1, 2, 3, 5 only.
- d. 3 only.
- e. 4 only.

Quiz navigation

Finish attempt ...

Time left **0:14:09**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **1**

Answer saved

Marked out of 1.00

Flag question

Both libraries and software frameworks are collections of reusable code. Both aim to provide solutions to common problems in Software development.

Select one:

- a. Agree
- b. Cannot say conclusively
- c. Strongly disagree
- d. The correct answer is not given
- e. Disagree

Next page

Quiz navigation

Finish attempt ...

Time left **0:12:39**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **3**

Answer saved

Marked out of 1.00

Flag question

What is/ are not an architectural constraint of REST?

1. Client-server architecture
2. Stateless
3. Cacheable
4. Strong Decoupling
5. Layered system

Select one:

- a. All of the above.
- b. 1, 2, 3 only.
- c. 1, 2, 3, 5 only.
- d. 3 only.
- e. 4 only.

Quiz navigation

Finish attempt ...

Time left **0:11:53**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **3**

Answer saved

Marked out of 1.00

Flag question

What is/ are not an architectural constraint of REST?

1. Client-server architecture
2. Stateless
3. Cacheable
4. Strong Decoupling
5. Layered system

Select one:

- a. All of the above.
- b. 1, 2, 3 only.
- c. 1, 2, 3, 5 only.
- d. 3 only.
- e. 4 only.

Quiz navigation

Finish attempt ...

Time left **0:11:29**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **4**

Answer saved

Marked out of 1.00

Flag question

What is NOT a drawback of Monolithic architecture?

Select one:

- a. Lack of flexibility regarding implementation technologies.
- b. High Global complexity.
- c. Can't scale individual components.
- d. High Local complexitity.
- e. The whole application instance should be replicated if there is a need to scale.

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:10:56**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **8**

Answer saved

Marked out of 1.00

Flag question

Which of the following is TRUE regarding a feature branch in git?

Select one or more:

- a. A feature branch is a copy of the main branch.
- b. Ideally, the main branch should never be directly modified. Instead, the changes should first be done in a feature branch.
- c. A Git workflow like the Github workflow encourages the creation of feature branches for every feature.
- d. None of them are true.
- e. A feature branch must always be created from the main branch.

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:08:44**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		

Question **9**

Answer saved

Marked out of 1.00

Flag question

What is the purpose of a version control system?

Select one:

a. To roll back changes to an earlier version.

b. To keep track of changes to code or other files.

c. To allow multiple people to work on the same file simultaneously.

d. None of the above.

e. All of the above.

Previous page

Next page

Quiz navigation

Finish attempt ...

Time left **0:08:17**

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30		